# GERMÁN KÜBER

## Senior Software Architect — .NET, Azure, Microservices & Distributed Systems

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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## PROFESSIONAL SUMMARY

Senior Software Architect with 15+ years designing and delivering production .NET / Azure systems for enterprise, regulated, and real-time domains, uniquely positioned for the AI-on-Azure modernization wave. Microsoft Certified: Azure Solutions Architect Expert (AZ-305) and Microsoft MVP for 8 consecutive years. Deep expertise in modern **.NET (8 / 9, ASP.NET Core, EF Core, .NET Aspire, Dapr)** and microservices architecture using Domain-Driven Design, CQRS, Event Sourcing, Saga patterns, Clean Architecture, and Vertical Slice Architecture, with production messaging across Azure Service Bus, RabbitMQ, Apache Kafka, MassTransit, and NServiceBus. Hands-on across the full Azure stack (AKS, API Management, Service Bus, Cosmos DB, Application Insights) with Terraform, Bicep, and ARM templates. Combines deep .NET / Azure architecture expertise with hands-on AI / LLM systems delivery — Azure OpenAI Service, multi-provider LLM orchestration, MCP (Model Context Protocol) servers and clients, and RAG pipelines on Azure AI Search. Remains active in **C# / .NET, TypeScript / Node.js, Python, and Rust** codebases while leading technical roadmap, ADRs, design reviews, and engineering teams. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

## CORE EXPERTISE

**Software Architecture & Patterns:** Domain-Driven Design (DDD) — tactical (entities, aggregates, value objects, repositories, domain events) and strategic (bounded contexts, context maps), CQRS, Event Sourcing, Saga patterns (orchestration and choreography), Clean Architecture / Hexagonal, Vertical Slice Architecture, SOLID principles, microservices design, event-driven architecture, polyglot persistence, eventual consistency, ADRs, C4 Model, design reviews

**Programming Paradigms & Design:** Object-Oriented Programming (OOP), Functional Programming, design patterns (Gang of Four, enterprise), architecture patterns, data structures, algorithm design, clean code

**Microsoft Stack & .NET:** .NET 6 / 7 / 8 / 9, C#, ASP.NET Core (Web API, MVC, Blazor), Entity Framework Core, .NET Aspire (cloud-native orchestration), Dapr (distributed application runtime), MediatR, AutoMapper, FluentValidation, MassTransit, NServiceBus, Polly (resilience), SignalR, gRPC, Dapper

**Microsoft Azure:** Azure Solutions Architect Expert (AZ-305 certified). App Service, Azure Functions, AKS (Azure Kubernetes Service), Azure Container Apps, Azure Service Bus, Event Grid, Event Hubs, Azure SQL Database, Cosmos DB, Azure Storage (Blob, Queue, Table), API Management, Azure Front Door, Entra ID (Azure AD), Key Vault, Application Insights, Azure Monitor, Azure DevOps, Logic Apps, Azure AI Search, Microsoft Fabric

**Messaging & Distributed Systems:** Azure Service Bus, RabbitMQ, Apache Kafka, MassTransit, NServiceBus, AMQP, MQTT, async runtimes (tokio in Rust, async / await in .NET), eventual consistency, distributed transactions, idempotency, outbox pattern, saga orchestration, retry policies with Polly

**AI / LLM Systems on Azure:** Azure OpenAI Service, multi-provider LLM orchestration (OpenAI, Anthropic, Gemini, AWS Bedrock, Vertex AI), MCP (Model Context Protocol) servers and clients, RAG pipelines on Azure AI Search, LangChain, LangGraph, multi-agent architecture, vector databases (Pinecone, Chroma, pgvector, Redis), LangSmith, RAGAS, DeepEval, OpenTelemetry GenAI semantic conventions

**APIs & Integration:** REST APIs, gRPC, GraphQL, API Gateways (YARP, Azure API Management, Kong, Nginx), OpenAPI / Swagger, OData, OAuth 2.0 / OIDC, IdentityServer / Duende, JWT, mTLS

**DevOps, IaC & Cloud-Native:** Docker, Kubernetes (AKS), Helm, CI / CD (Azure DevOps, GitHub Actions), Terraform, Bicep, ARM templates, OpenTelemetry, Prometheus, Grafana, Application Insights, distributed tracing, GitOps, cloud cost optimization

**Databases & Storage:** SQL Server / Azure SQL, PostgreSQL, MongoDB, Cosmos DB, Redis (cache and RediSearch), pgvector, polyglot persistence, query optimization, schema design for high-throughput systems

**Security & Identity:** OAuth 2.0 / OIDC, Entra ID (Azure AD), Microsoft Graph API, IdentityServer / Duende, Azure Key Vault, mTLS, API security (rate limiting, JWT, scopes), Zero Trust principles, OWASP Top 10

**Frontend (Full-Stack contexts):** Angular, React, TypeScript, Node.js, Blazor, reusable component libraries, UI architecture

**Engineering Practices & Leadership:** Test-Driven Development (TDD), pair programming, code review, technical roadmap, architectural decision records (ADRs), design reviews, mentoring, hiring and technical interviewing, cross-functional collaboration, internal training programs, on-call and incident response, legacy modernization and cloud migration

## PROFESSIONAL EXPERIENCE

### Lead Architect & Co-Founder

*KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain*

- Architected an end-to-end polyglot platform (Node.js / TypeScript, Python, .NET, Kubernetes, AWS); owned ADRs, service boundaries, technology selection, and deployment patterns as the senior-most engineer.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency on audio pipelines through careful design of async boundaries and message flow.
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Designed event-driven microservices architecture with multi-provider integration across OpenAI, Anthropic, Google Gemini, AWS Bedrock, Azure OpenAI Service, and Google Vertex AI, applying dynamic routing to pick the best model per task and cut inference spend.
- Integrated Azure OpenAI Service for production LLM workloads and designed RAG pipelines combining vector retrieval with Azure AI Search semantic ranking for enterprise-grade context retrieval.
- Implemented polyglot persistence across Pinecone, Chroma, pgvector, and Redis based on access patterns, query requirements, and consistency needs.
- Implemented MCP (Model Context Protocol) servers and clients to standardize tool access and context integration across distributed LLM workflows.
- Designed a Zero-Knowledge proof system using Noir, Barretenberg, and Garaga on StarkNet for verifiable expert credentials in a decentralized evaluation marketplace.
- Established observability stack with OpenTelemetry GenAI semantic conventions, LangSmith, and Application Insights for distributed tracing across services.
- Hired and led a distributed team of up to 3 engineers; owned technical interviews, architectural reviews, and code review standards.

### Fractional CTO (Part-time, hands-on)

*Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote*

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET (ASP.NET Core, Entity Framework Core, SQL Server), redesigning the architecture around Clean Architecture and microservices boundaries for long-term scalability.
- Designed and rolled out .NET architectural standards across the team — solution structure, layering, dependency injection patterns, error handling, validation pipelines (FluentValidation, MediatR) and integration testing practices.
- Migrated the real-time LLM service from OpenAI to Azure AI Foundry for compliance and data retention, cutting latency and improving scalability; introduced Spec-Driven Development (SDD).
- Architected AI-powered features combining .NET backend services with Node.js / TypeScript and Python (FastAPI, LangChain) AI orchestration, establishing the team's first LLM evaluation and observability practices.
- Optimized cloud infrastructure and LLM usage through semantic caching, prompt compression, and dynamic model selection, cutting combined API and cloud spend by 45%.

### Lead Software Engineer (Distributed Systems)

*Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote*

- Lead engineer for low-latency distributed systems in Rust (tokio, async) and Python / Cython, designing event-driven pipelines processing 10K+ opportunities per day from $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Designed and built a production trading bot in .NET (C#), integrating with Rust execution components and Python signal layers; owned the .NET architecture, deployment, and operational lifecycle in parallel with the broader distributed systems stack.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Owned 24 / 7 production reliability and incident response, including on-call rotation, post-mortems, and observability ownership with Prometheus, OpenTelemetry, and custom telemetry tracking 50+ real-time metrics.
- Designed high-throughput data pipelines and real-time signal processing across multi-chain blockchain ecosystems (EVM, Solana), with custom transaction delivery infrastructure achieving sub-slot inclusion (~400ms); reverse-engineered Geth and Reth, contributing to the REVM community.
- Acted as technical lead across the blockchain team, which grew from 4 to 10 engineers — code review, mentoring, technical interviewing, and architectural decisions.

### Senior Software Architect

*Paramo Technology — Oct 2020 – Sep 2021 — Remote*

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed event-driven microservices in .NET (C#, ASP.NET Core) handling real-time game state, wallet operations, and regulatory reporting; migrated relational databases to MongoDB where it made sense.
- Applied object-oriented design, Domain-Driven Design (DDD) — aggregates, value objects, domain events — design patterns, and CQRS for read/write separation across complex gambling and wallet domains.
- Implemented async messaging and event-driven workflows for inter-service communication with eventual consistency patterns across bounded contexts.
- Implemented distributed tracing across all services (OpenTelemetry, Application Insights, Datadog) to catch bugs early and resolve production issues faster; brought in Kubernetes to scale and built automated deploy pipelines.
- Led part of the hiring and designed the technical entry exam; created and delivered internal training programs on architecture patterns, design patterns, testing discipline, and operational best practices.

**Software Architect**

*CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)*

- Was the company's first employee; led architecture decisions across several enterprise product lines, designing microservices backends in .NET (C#, ASP.NET) using Domain-Driven Design with strategic and tactical patterns.
- Applied Clean Architecture and Hexagonal patterns for service boundaries, ensuring testability, evolvability, and decoupling of business logic from infrastructure concerns.
- Built reusable Angular and React frontends with component libraries shared across enterprise product lines, working with TypeScript / Node.js tooling and modern UI architecture patterns.
- Designed integration patterns and API contracts across multiple products and external systems, including REST APIs, async messaging, and service contracts.
- Delivered technical training programs and conference talks to partners and internal teams on architecture patterns, DDD, and software design.
- Collaborated directly with clients on-site for requirements gathering, architecture workshops, and technical adoption.

**Software Architect (Angular / .NET)**

*Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina*

- Designed and shipped full-stack features in Angular and .NET for internal hotel management platforms across booking, inventory, and operations, first as Team Lead and later as architect.
- Led the migration from AngularJS to Angular 2, modernizing component architecture and TypeScript adoption across the codebase.
- Redesigned the data model toward an event-oriented architecture, reducing coupling across integrations and improving system maintainability.
- Defined coding standards, code review practices, and architectural patterns for the engineering team.

**.NET Web Developer**

*Plus54 — 2011 – 2014 — Buenos Aires (earlier: Acsys 2009–2011, IBM 2008–2009)*

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

## KEY ACHIEVEMENTS

**Microsoft MVP, Developer Technologies** — 8 consecutive years (2017–present), recognized for technical content and community contributions across the .NET and Microsoft ecosystem.

**Microsoft Certified: Azure Solutions Architect Expert (AZ-305)** — Senior-level Azure architecture credential covering identity, governance, monitoring, data, business continuity, and infrastructure design.

**Net-Baires Founder** — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.

**80+ International Technical Talks** — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.

**Distributed Systems Engineering at Borderless** — Designed low-latency Rust / Python distributed systems processing $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms.

**Polkadot Builder Program Graduate** — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).

**Microsoft Student Partner** — 5 years (2016–2021).

## EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft

Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft

Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)

Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

## OPEN SOURCE & COMMUNITY LEADERSHIP

**Net-Baires (.NET Community)** — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.

**Patrones Para Mortales** — Founder (2020–present). Educational content brand on software design patterns and architecture.

**Crypto Para Mortales** — Founder (2023–present). Spanish-language blockchain and crypto education conference.

**YouTube channel (youtube.com/@germankuber)** — technical deep-dives on .NET, Azure, software architecture, Rust, blockchain, and AI.

## SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams